

Creation Date 26-Sep-2009

Revision Date 25-Mar-2024

Revision Number 5

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	Acrylic acid
Cat No. :	L04280
Synonyms	Acrylic acid, inhibited; 2-Propenoic acid; Acroleic acid
Index No	607-061-00-8
CAS No	79-10-7
EC No	201-177-9
Molecular Formula	C3 H4 O2
REACH registration number	-

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address

tech@alfa.com
www.alfa.com
Product safety Tel + +049 (0) 7275 988687-0

1.4. Emergency telephone number

Carechem 24: **+44 (0) 1235 239 670** (Multi-language emergency number)
Poison Information Center Mainz
www.giftinfo.uni-mainz.de
Telephone: +49(0)6131/19240

Exclusively for customers in Austria:
Poison Information Center (VIZ)
Emergency call 0-24 clock: **+43 1 406 43 43**
Office hours: Monday to Friday, 8am to 4pm, tel: +43 1 406 68 98

For customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

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SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable liquids

Category 3 (H226)

Health hazards

Acute oral toxicity

Category 4 (H302)

Acute dermal toxicity

Category 4 (H312)

Acute Inhalation Toxicity - Vapors

Category 4 (H332)

Skin Corrosion/Irritation

Category 1 A (H314)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

Specific target organ toxicity - (single exposure)

Category 3 (H335)

Environmental hazards

Acute aquatic toxicity

Category 1 (H400)

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H226 - Flammable liquid and vapor

H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled

H314 - Causes severe skin burns and eye damage

H335 - May cause respiratory irritation

H400 - Very toxic to aquatic life

Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P302 + P350 - IF ON SKIN: Gently wash with plenty of soap and water

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

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2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Stench

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Acrylic acid	79-10-7	EEC No. 201-177-9	>95	Flam. Liq. 3 (H226) Acute Tox. 4 (H302) Acute Tox. 4 (H312) Acute Tox. 4 (H332) Skin Corr. 1A (H314) Eye Dam. 1 (H318) STOT SE 3 (H335) Aquatic Acute 1 (H400)
4-Methoxyphenol	150-76-5	EEC No. 205-769-8	0.018-0.022	Acute Tox. 4 (H302) Skin Sens. 1 (H317) Eye Irrit. 2 (H319)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Acrylic acid	STOT SE 3 (H335) :: C>=1%	1	-

REACH registration number	-
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Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.
Skin Contact	Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.
Ingestion	Do NOT induce vomiting. Call a physician or poison control center immediately.
Inhalation	Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

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Difficulty in breathing. Causes burns by all exposure routes. . Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting; Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Carbon dioxide (CO₂). Dry chemical. Water mist may be used to cool closed containers. Chemical foam. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons

No information available.

5.2. Special hazards arising from the substance or mixture

Flammable. Corrosive material. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition. Vapors may form explosive mixtures with air. Do not allow run-off from fire-fighting to enter drains or water courses.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂).

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Wear self-contained breathing apparatus and protective suit. Evacuate personnel to safe areas. Remove all sources of ignition. Do not get in eyes, on skin, or on clothing. Take precautionary measures against static discharges.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage.

6.3. Methods and material for containment and cleaning up

Wear self-contained breathing apparatus and protective suit. Remove all sources of ignition. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Use spark-proof tools and explosion-proof equipment.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

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SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Use spark-proof tools and explosion-proof equipment. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges. Use only non-sparking tools.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep at temperatures between 15 ° and 25 °C. Keep away from heat, sparks and flame. Store indoors. May form explosive peroxides. Regularly check inhibitor levels to maintain peroxide levels below 1%. Keep container tightly closed in a dry and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Class 3

Switzerland - Storage of hazardous substances

Storage class - SC 3
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

Component	European Union	The United Kingdom	France	Belgium	Spain
Acrylic acid	TWA: 29 mg/m ³ (8h) TWA: 10 ppm (8h) STEL: 59 mg/m ³ (15min) STEL: 20 ppm (15min)	STEL: 20 ppm 15 min STEL: 59 mg/m ³ 15 min TWA: 10 ppm 8 hr TWA: 29 mg/m ³ 8 hr	TWA / VME: 10 ppm (8 heures). TWA / VME: 29 mg/m ³ (8 heures). STEL / VLCT: 20 ppm. indicative limit STEL / VLCT: 59 mg/m ³ . indicative limit	TWA: 2 ppm 8 uren TWA: 6.0 mg/m ³ 8 uren STEL: 20 ppm 15 minuten STEL: 59 mg/m ³ 15 minuten Huid	STEL / VLA-EC: 20 ppm (15 minutos). STEL / VLA-EC: 59 mg/m ³ (15 minutos). TWA / VLA-ED: 10 ppm (8 horas) TWA / VLA-ED: 29 mg/m ³ (8 horas) Piel
4-Methoxyphenol			TWA / VME: 5 mg/m ³ (8 heures).	TWA: 5 mg/m ³ 8 uren	TWA / VLA-ED: 5 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Acrylic acid	TWA: 29 ppm 8 ore.	TWA: 10 ppm (8)	STEL: 59 mg/m ³ 15	STEL: 59 mg/m ³ 15	TWA: 2 ppm 8 tunteina

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	Time Weighted Average TWA: 10 mg/m ³ 8 ore. Time Weighted Average STEL: 59 ppm 15 minuti. Short-term refers to a 1 minute reference exposure period STEL: 20 mg/m ³ 15 minuti. Short-term Pelle	Stunden). AGW - exposure factor 1 TWA: 30 mg/m ³ (8 Stunden). AGW - exposure factor 1 TWA: 10 ppm (8 Stunden). MAK TWA: 30 mg/m ³ (8 Stunden). MAK Höhepunkt: 10 ppm Höhepunkt: 30 mg/m ³	minutos STEL: 20 ppm 15 minutos TWA: 10 ppm 8 horas TWA: 29 mg/m ³ 8 horas Pele	minuten TWA: 29 mg/m ³ 8 uren	TWA: 6 mg/m ³ 8 tunteina Ceiling: 15 ppm Ceiling: 45 mg/m ³
4-Methoxyphenol			TWA: 5 mg/m ³ 8 horas		

Component	Austria	Denmark	Switzerland	Poland	Norway
Acrylic acid	MAK-KZGW: 20 ppm 15 Minuten MAK-KZGW: 59 mg/m ³ 15 Minuten MAK-TMW: 10 ppm 8 Stunden MAK-TMW: 29 mg/m ³ 8 Stunden	TWA: 2 ppm 8 timer TWA: 5.9 mg/m ³ 8 timer STEL: 20 ppm 15 minutter STEL: 59 mg/m ³ 15 minutter Hud	STEL: 20 ppm 15 Minuten STEL: 59 mg/m ³ 15 Minuten TWA: 10 ppm 8 Stunden TWA: 29 mg/m ³ 8 Stunden	STEL: 29.5 mg/m ³ 15 minutach TWA: 10 mg/m ³ 8 godzinach	TWA: 10 ppm 8 timer TWA: 29 mg/m ³ 8 timer STEL: 20 ppm 15 minutter. value from the regulation STEL: 59 mg/m ³ 15 minutter. value from the regulation
4-Methoxyphenol	MAK-KZGW: 10 mg/m ³ 15 Minuten MAK-TMW: 5 mg/m ³ 8 Stunden	TWA: 5 mg/m ³ 8 timer STEL: 10 mg/m ³ 15 minutter		TWA: 5 mg/m ³ 8 godzinach	TWA: 5 mg/m ³ 8 timer STEL: 10 mg/m ³ 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Acrylic acid	TWA: 29 mg/m ³ TWA: 10 ppm STEL : 59 mg/m ³ STEL : 20 ppm	TWA-GVI: 10 ppm 8 satima. TWA-GVI: 29 mg/m ³ 8 satima. STEL-KGVI: 20 ppm 15 minutama. applies for a reference period of 1 minute STEL-KGVI: 59 mg/m ³ 15 minutama. applies for a reference period of 1 minute	TWA: 10 ppm 8 hr. TWA: 29 mg/m ³ 8 hr. STEL: 20 ppm 15 min STEL: 59 mg/m ³ 15 min	STEL: 59 mg/m ³ STEL: 20 ppm TWA: 29 mg/m ³ TWA: 10 ppm	TWA: 30 mg/m ³ 8 hodinách. Ceiling: 60 mg/m ³
4-Methoxyphenol			TWA: 5 mg/m ³ 8 hr. STEL: 15 mg/m ³ 15 min		

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Acrylic acid	TWA: 10 ppm 8 tundides. TWA: 29 mg/m ³ 8 tundides. STEL: 20 ppm 15 minutites. STEL: 59 mg/m ³ 15 minutites.	TWA: 29 mg/m ³ 8 hr TWA: 10 ppm 8 hr STEL: 59 mg/m ³ 15 min reference period of 1 minute STEL: 20 ppm 15 min reference period of 1 minute	STEL: 20 ppm STEL: 59 mg/m ³ TWA: 10 ppm TWA: 29 mg/m ³	STEL: 59 mg/m ³ 15 percekben. CK Refers to a reference time of 1 minute TWA: 29 mg/m ³ 8 órában. AK	TWA: 2 ppm 8 klukkustundum. TWA: 5.9 mg/m ³ 8 klukkustundum. Skin notation Ceiling: 4 ppm Ceiling: 11.8 mg/m ³
4-Methoxyphenol			TWA: 5 mg/m ³		TWA: 5 mg/m ³ 8 klukkustundum. Ceiling: 10 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Acrylic acid	STEL: 59 mg/m ³ STEL: 20 ppm TWA: 5 mg/m ³ TWA: 1.7 ppm	Ceiling: 59 mg/m ³ Ceiling: 20 ppm TWA: 10 ppm IPRD TWA: 29 mg/m ³ IPRD	TWA: 29 mg/m ³ 8 Stunden TWA: 10 ppm 8 Stunden STEL: 59 mg/m ³ 15 Minuten STEL: 20 ppm 15 Minuten	TWA: 10 ppm TWA: 29 mg/m ³ STEL: 20 ppm 15 minuti STEL: 59 mg/m ³ 15 minuti	TWA: 10 ppm 8 ore TWA: 29 mg/m ³ 8 ore STEL: 20 ppm 15 minute STEL: 59 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Acrylic acid	TWA: 5 mg/m ³ 1775	Ceiling: 59 mg/m ³	TWA: 29 mg/m ³ 8 urah	Binding STEL: 20 ppm	

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	MAC: 15 mg/m ³	TWA: 10 ppm TWA: 29 mg/m ³	TWA: 10 ppm 8 urah Koža STEL: 20 ppm 15 minutah applies for a reference period of 1 minute STEL: 59 mg/m ³ 15 minutah applies for a reference period of 1 minute	15 minuter Binding STEL: 59 mg/m ³ 15 minuter TLV: 10 ppm 8 timmar. NGV TLV: 29 mg/m ³ 8 timmar. NGV	
4-Methoxyphenol	MAC: 0.5 mg/m ³		TWA: 5 mg/m ³ 8 urah		

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

Workers; See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Acrylic acid 79-10-7 (>95)	DNEL = 1mg/cm ²		DNEL = 1mg/cm ²	

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Acrylic acid 79-10-7 (>95)	DNEL = 30mg/m ³	DNEL = 30mg/m ³	DNEL = 30mg/m ³	DNEL = 30mg/m ³
4-Methoxyphenol 150-76-5 (0.018-0.022)				DNEL = 3mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Acrylic acid 79-10-7 (>95)	PNEC = 0.003mg/L	PNEC = 0.0236mg/kg sediment dw	PNEC = 0.0013mg/L	PNEC = 0.9mg/L	PNEC = 1mg/kg soil dw
4-Methoxyphenol 150-76-5 (0.018-0.022)	PNEC = 0.0136mg/L	PNEC = 0.125mg/kg sediment dw		PNEC = 10mg/L	PNEC = 0.017mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
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Acrylic acid 79-10-7 (>95)	PNEC = 0.0003mg/L	PNEC = 0.002346mg/kg sediment dw		PNEC = 0.03g/kg food	
4-Methoxyphenol 150-76-5 (0.018-0.022)	PNEC = 0.00136mg/L	PNEC = 0.0125mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Wear safety glasses with side shields (or goggles) Goggles (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber	See manufacturers	-	EN 374	(minimum requirement)
Neoprene	recommendations			
Natural rubber				
PVC				

Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Organic gases and vapours filter Type A Brown conforming to EN14387

Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State

Liquid

Appearance Odor

Colorless
Stench

ALFAAL04280

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Odor Threshold	No data available	
Melting Point/Range	13 °C / 55.4 °F	
Softening Point	No data available	
Boiling Point/Range	139 °C / 282.2 °F	@ 760 mmHg
Flammability (liquid)	Flammable	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	Lower 2 Vol%	
	Upper 15.9 Vol%	
Flash Point	48 °C / 118.4 °F	Method - No information available
Autoignition Temperature	374 °C / 705.2 °F	
Decomposition Temperature	No data available	
pH	1.0-2	
Viscosity	1.3 mPa s at 20 °C	
Water Solubility	Miscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Acrylic acid	0.46	
4-Methoxyphenol	1.3	
Vapor Pressure	@ 3.8 mbar °C 20	
Density / Specific Gravity	1.050	
Bulk Density	Not applicable	Liquid
Vapor Density	2.48 (Air = 1.0)	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula	C3 H4 O2
Molecular Weight	72.06
Explosive Properties	explosive air/vapour mixtures possible
Self-accelerating polymerisation temperature (SAPT)	55 -75 °C (all packages) Heat of Polymerization (kJ/kg) = 1079

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

Yes

10.2. Chemical stability

Hazardous polymerization may occur. May form explosive peroxides on prolonged storage.
Hygroscopic.

10.3. Possibility of hazardous reactions

Hazardous Polymerization	Hazardous polymerization may occur upon depletion of inhibitor.
Hazardous Reactions	No information available.

10.4. Conditions to avoid

Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to light. Exposure to moist air or water. Incompatible products.

10.5. Incompatible materials

Strong oxidizing agents. Strong bases. oxygen. Peroxides. Halogens. Aldehydes. Amines.
Acid anhydrides.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO₂).

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SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information Harmful by inhalation, in contact with skin and if swallowed

(a) acute toxicity;

Oral Category 4
Dermal Category 4
Inhalation Category 4

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Acrylic acid	468-1500 mg/kg (Rat)	>2000 mg/kg (Rabbit)	>5.1 mg/L/4h (Rat)
4-Methoxyphenol	1600 mg/kg (Rat)	LD50 > 2000 mg/kg (Rabbit)	-

(b) skin corrosion/irritation; Category 1 A

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory Based on available data, the classification criteria are not met
Skin Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met
Reproductive Effects Experiments have shown reproductive toxicity effects on laboratory animals.

(h) STOT-single exposure; Category 3
Results / Target organs Respiratory system.

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met
Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Other Adverse Effects See actual entry in RTECS for complete information

Symptoms / effects, both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any

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known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Acrylic acid	LC50: = 222 mg/L, 96h semi-static (Brachydanio rerio)	EC50: = 95 mg/L, 48h (Daphnia magna)	EC50: = 0.17 mg/L, 96h (Pseudokirchneriella subcapitata) EC50: = 0.04 mg/L, 72h (Desmodesmus subspicatus)
4-Methoxyphenol	LC50: = 28.5 mg/L, 96h flow-through (Oncorhynchus mykiss) LC50: = 84.3 mg/L, 96h flow-through (Pimephales promelas)		

Component	Microtox	M-Factor
Acrylic acid		1
4-Methoxyphenol	EC50 = 3.66 mg/L 5 min EC50 = 4.30 mg/L 15 min EC50 = 4.61 mg/L 30 min	

12.2. Persistence and degradability

Persistence

Degradation in sewage treatment plant

Readily biodegradable

Miscible with water, Persistence is unlikely, based on information available.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Acrylic acid	0.46	No data available
4-Methoxyphenol	1.3	No data available

12.4. Mobility in soil

The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

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13.1. Waste treatment methods

Waste from Residues/Unused Products	Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.
Switzerland - Waste Ordinance	Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600 https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number	UN2218
14.2. UN proper shipping name	ACRYLIC ACID, STABILIZED
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	3
14.4. Packing group	II

ADR

14.1. UN number	UN2218
14.2. UN proper shipping name	ACRYLIC ACID, STABILIZED
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	3
14.4. Packing group	II

IATA

14.1. UN number	UN2218
14.2. UN proper shipping name	ACRYLIC ACID, STABILIZED
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	3
14.4. Packing group	II

14.5. Environmental hazards	Dangerous for the environment Product is a marine pollutant according to the criteria set by IMDG/IMO
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14.6. Special precautions for user	Storage conditions in Section 7 should also be met during transportation. Inhibitors have been added to stabilize this product. Inhibitor levels should be maintained. Hazardous polymerization may occur upon depletion of inhibitor.
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14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods
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SECTION 15: REGULATORY INFORMATION

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15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Acrylic acid	79-10-7	201-177-9	-	-	X	X	KE-29442	X	X
4-Methoxyphenol	150-76-5	205-769-8	-	-	X	X	KE-23353	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Acrylic acid	79-10-7	X	ACTIVE	X	-	X	X	X
4-Methoxyphenol	150-76-5	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Acrylic acid	79-10-7	-	Use restricted. See item 75. (see link for restriction details)	-
4-Methoxyphenol	150-76-5	-	Use restricted. See item 75. (see link for restriction details)	-

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Acrylic acid	79-10-7	Not applicable	Not applicable
4-Methoxyphenol	150-76-5	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

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National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Acrylic acid	WGK 2	Class I : 20 mg/m ³ (Massenkonzentration)
4-Methoxyphenol	WGK1	

Component	France - INRS (Tables of occupational diseases)
4-Methoxyphenol	Tableaux des maladies professionnelles (TMP) - RG 65

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Acrylic acid 79-10-7 (>95)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H226 - Flammable liquid and vapor
H302 - Harmful if swallowed
H312 - Harmful in contact with skin
H314 - Causes severe skin burns and eye damage
H317 - May cause an allergic skin reaction
H318 - Causes serious eye damage
H319 - Causes serious eye irritation
H332 - Harmful if inhaled
H335 - May cause respiratory irritation
H400 - Very toxic to aquatic life

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

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ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Training Advice

Chemical incident response training.

Prepared By

Health, Safety and Environmental Department

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26-Sep-2009

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25-Mar-2024

Revision Summary

SDS sections updated, 9, 14.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet